scientific reports

OPEN

# Reframing the filter bubble through diverse scale effects in online music consumption

Dougal Shakespeare[1,2]✉, Victor Chareyron[1] & Camille Roth[1,2]

**The impact of platforms' algorithmic curation devices on users' consumption diversity has long since been the subject of debate in popular culture and academia alike. In recent years, methods to concretely appraise this debate have taken a comparative turn, with emerging research strands seeking to assess the influence of algorithmic recommendations relative to users' *organic* behaviour (i.e. those with a relative lack of algorithmic influence). Nevertheless, as the contribution of this work clearly shows, contradictory findings that algorithmic devices may both foster "bubbly" confinement dynamics and diverse exploration appear to stem, at least in part, from tensions surrounding information representation and scale. Taking into account various scales, affordances, and diversity measures, the contribution of this work conducts an empirical analysis of approximately 50,000 Deezer users' artist consumption histories across different affordances offering various degrees of algorithmic assistance. Transitioning through three scales of increasing aggregation—intra-sessions, inter-sessions and inter-affordances—our work underscores how shifts in perspective, particularly in information representation, can yield markedly different conclusions concerning the role of algorithmic devices in fostering diversity. However, when considered collectively, these perspectives coalesce to provide a nuanced update to the seminal filter bubble narrative: algorithmic curation devices may introduce more novelty than what users' achieve organically, yet concurrently, this novelty is more semantically confined. Consequently, our research significantly contributes towards an enriched understanding of the intricate dynamics surrounding on-platform algorithmic devices.**

More and more information access is mediated by digital platforms and the various affordances they embody. Now ubiquitous personalised *algorithmic* curation devices (*A*) play a key role in this process, affecting what, how and when users are exposed to information on-platform. Even though some studies appear to confirm the seminal hypothesis that these devices may have a constraining effect upon diversity, isolating users in so-called filter bubbles[1] of overly personalised, homogeneous content[2,3], an array of prior works appear to suggest that they rather have the potential to increase consumption diversity[4–16]. Notwithstanding, empirically testing the validity of such claims frequently requires defining a suitable baseline reflecting users' *organic* (*O*) consumptions, that is, those performed with a relative lack of algorithmic assistance. The type of usage traces utilised to this end depends largely on the type of diversity appraisal being conducted. Broadly, this can be classified under two strands of literature based on whether they appraise concentration dynamics in terms of *A consumption vs. A exposure*[4,5,12,13], or in terms of *A consumption vs. another lesser A affordance*[2,8–11,16–18].

The former is most commonly conducted within the field of Political Science and is driven by the theory of selective exposure, a core concept in communication and media studies which describes the propensity for individuals to consume media that aligns with their prior beliefs[19] thereby limiting exposure to cross-cutting content. Empirical works in this direction typically explore how alignment differs within the information sets algorithmic devices expose users to, and then, downstream, within the information sets that users' organically select[4,5]. Under this framing, alignment is thus operationalized by the extent to which users' consumption and exposure logs in the algorithmic affordance remain more or less confined within users' prior preference bubbles[5]. Whilst by design, algorithmic devices tend to generate recommendations that align with and thus, are *within* users' prior preference bubbles recent empirical results suggest that perhaps surprisingly, users' own organic selections of algorithmic recommendations can further constrain their consumption diversity[4,5,12,15] sometimes, to an even greater extent[4,5].

The second strand of literature seeks to appraise algorithmic confinement by comparing users' consumption diversity across *different* affordances which each offer variable levels of algorithmic assistance. Following Davis formulation[20], an affordance describes *how* an object enables and constrains; in the context of online platforms,

[1]CAMS, Centre d'Analyse et de Mathématique Sociale, CNRS/EHESS, Paris, France. [2]Centre Marc Bloch, Computational Social Science Team, Berlin, Germany. ✉email: dougal.shakespeare@ehess.fr

this generally corresponds to one way of accessing content or information. For instance, on music streaming platforms which are also the focus of many studies within this strand this includes organic personal playlist, algorithmic recommendation devices and human editorial curations to name a few[2,8–10,16,17]. While most existing works focus on a variety of affordances within one platform, some perform cross-platform comparisons whereby each platform is also deemed an affordance in itself[14,18]. Put simply, the diversity of the content consumed within an affordance would be compared with that of another affordance mostly empirical however sometimes hypothesised[11]. Typically, this involves comparing users' consumptions within *A* and *O* affordances. It can also more broadly be conceptualised in terms of comparing consumptions within an algorithmic affordance to a lesser algorithmic one. Like several of the previously detailed *A* exposure works, existing research generally concludes that users' algorithmic consumptions are consistently either less[9,16] or no less concentrated[8] than their organic ones.

Whilst both branches of research appear to imply that confinement dynamics may be amplified by self-imposed organic personalisation a handful of research adds further nuance to this narrative regarding how concentration dynamics vary for specific content[21], user categories[9] and, perhaps most importantly, when relying on diversity measures which leverage more geometric feature representations[2,13,17]. The latter hints at an important dichotomy regarding how information and, therefore, diversity, is being constructed: that is, with *discrete* or *dense* representations. Indeed, existing research tends to rely on discrete information representations built upon either item categories (e.g., left- vs. right-wing, music genres, ect.) or simply items themselves. Under this framing diversity is typically appraised through *novelty*, be that in terms of users' exploration vs. exploitation dynamics (i.e., exploring new items / exploiting known ones) or repetition-based measures whose methods rely on set-theoretic or entropy based approaches. In contrast, recent works[2] have also sought to study diversity with more dense, continuous item representations, essentially capturing users' diversity through the *spatial disparity* of the items they consume. Under this framing, research suggests that algorithmic devices may in fact have a constraining effect upon users' diversity — Anderson et al.[2] show that users' *A* consumptions are consistently less diverse than their *O* ones on the Spotify platform. These approaches highlight a potential tension which resides between diversity measures which rely on *discrete* or *dense* information representations and, thus, diversity measures which capture *novelty* or *spatial* disparity. This is in line with the recent findings of Mok et al.[17] which suggest that at the user level, there is little connection between the two perspectives.

Nevertheless, Mok et al. do not disaggregate affordances when drawing these conclusions and furthermore, an attempt to compare their results to the affordance-centric perspective runs into the issue of scale variance — Mok et al. work at the short-term (per week) and long-term (4-year scale) whilst Anderson et al. work at the aggregate 1-month scale[2]. This hints at another factor which may have a drastic effect upon the extent to which research may or may not support the "filter bubble" hypothesis: the temporal or user aggregation scale applied. Existing research typically groups effects for entire platforms[10,11,14], user groups[13,22] or whole user consumption histories[9,15] and thus, works at a broad aggregate scale. Whilst some studies distinguish short- vs. longer-term effects[17,18], little further refinement is commonly performed to segment consumption logs into temporally consistent units. In particular, despite the increasing interest in so-called 'session-based recommender systems' in the literature dedicated to the conception of algorithms[23], empirical works leveraging specifically the level of sessions congruent with the bursty behaviour of users remain quite rare.

The contribution of this work seeks to empirically appraise the extent to which diversity methods built upon discrete or dense information representations capture heterogeneous perspectives of users' consumption diversity across *organic* and lesser *organic* affordances and furthermore, explore how this varies with aggregation scale. To this avail, we perform an empirical multi-affordance analysis of the listening histories of aproximately 50K users of the French based music streaming platform, Deezer. Leveraging short term *sessions* of continuous platform engagement we study diversity at 3 successive scales of increasing aggregation: (1) intra-sessions, (2) inter-sessions (the closest to existing aggregate analysis[2,8,9,16]) and (3) inter-affordances. Consequently, our work shows how confinement conclusions can drastically differ dependent on the scale and information representation chosen. However, when parallel perspectives are considered en masse they can in many cases coalesce to form a more nuanced narrative. This allows us to provide a substantial update of the classical filter bubble narrative which we hope will contribute towards a more enriched understanding of the intricate dynamics surrounding on-platform confinement: algorithmic curation devices on music steaming platforms may introduce more novelty than what users achieve organically, yet concurrently, this novelty is more spatially confined.

## The deezer platform

Our study falls into the strand of literature which seeks to compare users' consumption diversity between various on-platform affordances with different levels of algorithmic / organic assistance. To this end, we focus on the French-based music streaming platform Deezer which offers a fruitful enviroment due to the richness of on-platform affordances offered.

### Dataset scope and affordance classification

Prior to performing any concentration computations, our initial pre-filtered dataset consists of 1 month (March 2023) of listening histories from a random sample of 48, 243 anonymised Deezer users who streamed at least once in that month and pay for a subscription to the platform. Although not analysed directly, this user set is used to compute our user- and affordance-centric datasets which form the basis of our analysis. This is further clarified in the *Affordances and User Centric Datasets* section below. We pre-process our dataset by filtering listening events shorter than 30 seconds as these are deemed to be so-called 'skips' rather than streams[2]. After applying this data processing pipeline we work with 23, 944, 195 listening events covering 168, 685 unique artists. We then label each stream in accordance with the affordance adopted to instantiate it. More precisely, the Deezer platform keeps highly granular logs regarding where on platform a song was streamed, thus allowing us

to classify affordances in a more fine-grained manner than simply *organic* / "*programmed*" as outlined in prior works[2]. In particular, we divide organic affordances into two finer grained classes which are semantically and, as we shall shortly demonstrate, behaviourally distinct: playlists (*P*) vs. query-based (*Q*). Indeed, *P* affordances refer to playlists which have either been *curated* or *saved* organically by the user whilst *Q* affordances refer to all other modes of organic access which most commonly result from a user's direct queries to a platform's database (e.g. search result pages[24,25]) or browsing a platform's informational landscape with minimal algorithmic assistance (e.g. on an artist's profile page). Similarly, we are able to break down the "programmed" affordance further into algorithmic (*A*) and editorial (*E*) affordances based on the extent to which they rely on algorithmic or, rather, organic human curations from editors. *A* refers to the platform's plethora of Recommender Systems (e.g., the popular never-ending mix called 'Flow' on Deezer) whilst *E* affordances correspond to playlists curated by Deezer Editorial staff including playlists variously called "10s electronic", "Rock & Chill", etc. All things considered this leaves us with 4 affordances which can be used to classify the origins of users' streams, each of which offers a unique perspective regarding organic-algorithmic assistance: *organic playlists* (*P*), *organic query* (*Q*), *editorial curation* (*E*) and *algorithmic curation* (*A*).

### The session scale

Deezer, like many other platforms, offers much flexibility regarding how users' streams can be aggregated. Due to the bursty nature of users' platform access streams typically cluster into so-called *sessions* of continuous platform engagement thus, providing a temporally cohesive scale to work at. In our work we define a new session for a user when the inter-listening event time between any two consecutive streams exceeds 20 minutes – this captures 91% of inter-stream times and results in 2, 209, 591 sessions. To understand how users' affordance adoption preferences correspond to the session scale, we compute for each user the proportion of their sessions which are monads, dyads, triads or tetrads i.e., drawing on 1, 2, 3, or 4 distinct affordances (in any order). Surprisingly, the majority of users' sessions only make use of one affordance. On average 73.9% of a user's sessions are monadic and furthermore, more monadic users are more active: computing the percent of monadic sessions at the platform level we even observe an increased prevalence of this session type (83.3%). This simplifies our analysis, allowing us to focus exclusively on monadic sessions which, although not all, still make up the majority of users' platform interactions whilst preserving users' relative affordance preferences (see Supplementary Information).

Focusing from hereon out on analysing users' diversity using solely monadic sessions, we shall compute and compare diversity between affordances at 3 scales of increasing aggregation. Firstly, at the session scale by conducting an intra-session diversity analysis. Secondly at the inter-session scale (the closest to existing aggregate analysis[2,8,9,16]) treating sessions as cohesive units of measurement in their own right and then thirdly, we move to the affordance scale to consider inter-affordance similarity. These three stages of comparison form the groundwork of our analysis and provide various perspectives to study *diversity at scale*.

### Affordance and user centric datasets

We subset our monadic dataset by applying two filtering strategies. The first constructs, for each affordance, the subset of users and their corresponding sessions who show marked levels of engagement ($\geq 3$ monadic sessions) with the *P*, *Q*, *A* or *E* affordance. The second filtering strategy constructs a subset of users and their associated sessions who are multi-affordance that is, they display marked engagement ($\geq 3$ monadic sessions) with *each* of the 4 platform affordances. Ultimately, the former proves useful for an *affordance-centric* analysis whilst the latter facilitates a *user-centric* analysis which can be used to sense-check the conclusions derived at the affordance scale, thus reducing the risk of ecological fallacy. For further details we refer the reader to the "Methods" section.

Although not detailed here due to space constraints we remark that we were also able to reproduce each of the experiments on two novel affordance-centric and user-centric datasets derived from a random 50K user dataset for the month of March 2022. Affordance differences detailed are largely consistent across each of the datasets tested as described in the Supplementary Information section.

## Artist representation and diversity measurement approach

To explore the tensions between users' consumption diversity dynamics both in terms of *novelty* and *spatial disparity*[17] we construct both discrete and dense item representations upon which respectively novelty and spatial disparity-based diversity measurements can be performed. The former relies on set-theoretic approaches whilst the latter leverages a high-fidelity item embedding space. In both cases we work with artists rather than songs as items and thus, both spatial and novelty based diversity measures are computed either on artists representations in the former case or in the latter, directly on artists themselves.

### Spatial disparity measures

To capture users' consumption diversity we construct an artist-based embedding space by applying SVD to a popularity normalised artist-artist within-session co-occurrence matrix *M*. We choose to apply SVD over other more "state-of-the-art" embedding models since it is both more interpretable and reproducible[26]. Furthermore, it is worth highlighting that SVD has also been shown to be closely connected to the Skip-Gram Negative Sampling (SGNS) variant of the word2vec model[27]. As such, applying this method further connects our work to existing literature applying SGNS models to conduct consumption diversity studies on multi-affordance platforms[2,28]. We provide in Fig. 1 a visual two-dimensional projection of our SVD embedding space to give a sense of the artist space we are working with and its likely spikiness.

We compute the disparity a given session by computing an activity-weighted "Generalist-Specialist" (GS) score[2,28]. At a high level, this measure captures the expected cosine similarity between a session's centroid in vector space and the vector representation of a randomly drawn stream from within that session. Higher GS score indicate that a session is more concentrated in vector space and therefore, more *specialist* whereas lower

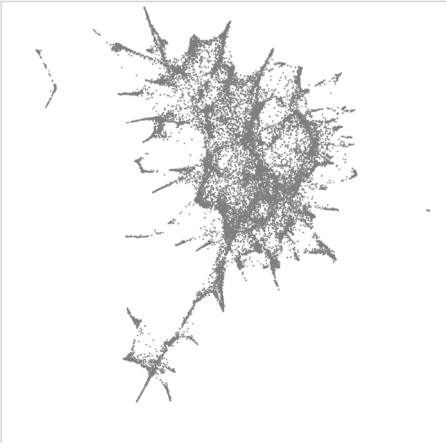

**Fig. 1**. PaCMAP[29] projection of the artist embedding space.

| affordance $F$ | $n_F$ | $\langle \mathrm{GS}(S) \rangle$ | $\langle \gamma \rangle$ | $\langle \mathrm{GS}(S_F) \rangle$ |
|---|---|---|---|---|
| $A$ | 12,411 | 0.829 ($\pm$0.073) | ***0.098** ($\pm$0.159) | 0.870 ($\pm$0.078) |
| $E$ | 10,923 | 0.874 ($\pm$0.081) | *0.130 ($\pm$0.222) | 0.852 ($\pm$0.105) |
| $P$ | 28,896 | ***0.809** ($\pm$0.083) | 0.244 ($\pm$0.255) | 0.860 ($\pm$0.079) |
| $Q$ | 36,004 | 0.919 ($\pm$0.065) | 0.134 ($\pm$0.210) | ***0.793** ($\pm$0.106) |

**Table 1**. Intra-session ($\mathrm{GS}(S)$) and inter-session ($\gamma$ and $\mathrm{GS}(S_F)$) concentrations for affordance-centric sub-datasets. Standard deviations are mentioned inside parentheses and averages are reported over all users who displayed marked engagement with the affordance. Bold values mark minimums (i.e. highest diversity) per affordance and * denotes that the diversity of this affordance is significantly greater than all other affordance distributions across several pairwise one-tailed Kolmogorov-Smirnov tests ($\alpha = 0.001$ and Bonferroni correction applied). $n_F$ denotes the number of users within each affordance group.

GS score indicate that a session is more spread in vector space and thus, more *generalist* and diverse. Since artists in our space tend to be clustered by genres, higher GS score associated with a session can likewise be interpreted as the sessions being more genre homogeneous and thus, semantically consistent whilst lower GS scores indicate that the session is generally, more genre diverse. For a more formal analysis of this relationship we refer readers to the Supplementary Information section. Similarly, it holds that a more spread, spatially diverse session with a lower GS score could be interpreted as also being more semantically diverse. This is due to the fact that so-called distributional semantic models such as SVD are known to capture structural paradigmatic semantics[30] within their geometry. As such, a session with a low GS score is likely to consist of artists which on average tend to have weak paradigmatic connections between them.

### Novelty measures

The relation between spatial disparity and novelty is non-trivial: typically a session can be spatially disparate but made of recurrent elements only and, conversely, a spatially compact session could comprise exclusively novel elements. Put simply, spatial disparity is highly dependent on how exactly users navigate a vector space but does not indicate necessarily whether they go to novel or already-visited places and thus, whether they repeat consumptions or not. To address this issue, we configure a novelty-based appraisal of diversity in set theoretic terms. More precisely, we capture users' novelty (1) through the repetition of items within a session (when studying diversity intra-sessions) and (2) through the overlap between consumed item sets (when studying diversity inter-sessions and inter-affordances). The former relies on a repetition-based formulation and is captured with a simple length-adjusted Redundancy measure, whilst the latter is more set-theoretic and is measured with the overlap measure ($\gamma$) which makes use of the Bipartite Configuration Model (BiCM)[31] that we detail further below and in the "Methods" section.

## Intra-session diversity

### Spatial disparity analysis

We first centre our attention on analysing the diversity of non-fandom sessions (i.e. sessions containing more than $> 1$ artist) in SVD space using both our affordance-centric and user-centric datasets. Starting with the former and analysing $\langle \mathrm{GS}(S) \rangle$ scores per affordance we observe marked variations in consumption diversity dependent on the affordance adopted. We present these results in Table 1 whereby an affordance having a significantly greater diversity than all other affordances is appraised by applying several pairwise one-tailed Kolmogorov-

| | Spatial disparity | Novelty |
|---|---|---|
| Intra-session | $Q$ —— $E$ —— $A - P$ | $Q$ —— $P - E$ —— $A$ |
| Inter-session | $A - P - E$ —— $Q$ | $P$ —— $Q - E - A$ |

**Table 2**. Summary of the findings for affordance hierarchy from the least to most diverse, from left to right (whereby line lengths are simply proportional to unidimensional distances between reported diversities in each case) for affordance-centric results, depending on intra- vs. inter-session scale, and on spatial disparity vs. novelty.

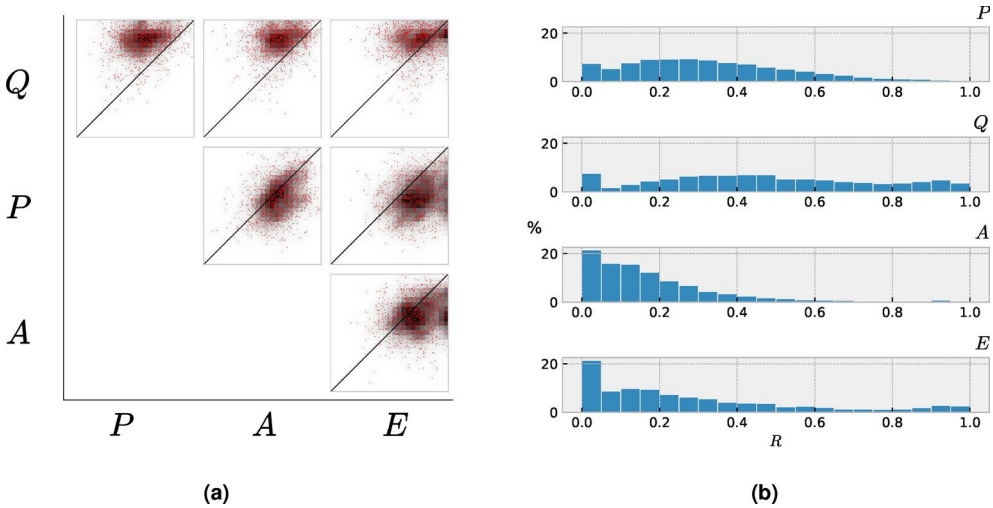


**Fig. 2**. Intra-session Diversity Analysis. Left (**a**) details distribution of users of the user-centric multi-affordance filtered dataset with respect to their *GS* scores for non-identical affordance tuples. Densities above (respectively, below) the line of equality indicate greater artist concentration i.e., lower diversity within sessions which utilise the affordance indicated in the grid rows (respectively, columns). Right (**b**) shows length adjusted intra-session Redundancy distributions computed for each affordance-centric user set corresponding to each of the 4 platform affordances.

Smirnov tests ($\alpha = 0.001$). To account for these multiple comparisons we correct $\alpha$ via Bonferroni correction. We chose to apply the Kolmogorov-Smirnov test since its non-parametric nature enables its application to non-normal distributions, which are particularly prevalent in our novelty-based inter-session analysis. As displayed in Table 1 and visually in Table 2, quite strikingly, there is a clear difference between the diversity of organic *P* and organic *Q* sessions, despite them previously being grouped under the more general *organic* umbrella term[2]. In fact, they even display polar opposite diversity dynamics at this scale. More precisely, organic *Q* and, then, *E* sessions appear to be the least diverse whilst algorithmic *A* sessions and even more so, organic *P* sessions tend to be the most diverse, according to their *GS* scores. Put simply, at this level algorithmic sessions do not seem to be more bubbly than organic affordances. We observe comparable results with the user-centric perspective: for each multi-affordance user we compute $\langle \mathrm{GS}(S) \rangle$ scores per each of the 4 affordances they engage with and then for each pairwise affordance combination we project users as points in *GS* space as depicted in Fig. 2a. Upon inspecting this figure it becomes clear that once again, users' *Q* sessions are consistently their least diverse whilst users' organic *P* sessions are, on the contrary, consistently their most diverse. In short, the diversity hierarchy observed during our previous affordance-centric analysis is found once again when considering differences between all affordances (see Supplementary Information).

## Novelty analysis

To measure intra-session diversity using discrete item representations, therefore capturing novelty, we first compute the proportion of previously removed fandom sessions which reside within each session type thus, allowing us to appraise the extent to which different affordances tend to foster at least *some* artist novelty. Without loss of generality let *S* denote a user's session which is, more formally, a list of streamed artist ids. Similarly, let $S_F$ denote the collection of monadic sessions which the user instantiated with affordance $F \in \{A, E, P, Q\}$. Thus, the proportion of fandom sessions within a given affordance is captured as:

$$\phi_F = \frac{|\{S \in S_F : |\operatorname{supp} S| = 1\}|}{|S_F|} \tag{1}$$

where $\operatorname{supp} S$ denotes the support of the session set *S* (i.e., the set of unique artist IDs reached in *S*). Focusing first, on our affordance-centric subsets we observe that, just as organic *Q* sessions were found to be the least diverse from a dense, spatial perspective, likewise are they the least diverse from a discrete, novelty-based perspective: they contain by far the most fandom sessions, with an average of 46.9% over users. To put things in perspective, on average 26.13%, 18.40% and 13.54% of a user's *E*, *P* and *A* sessions are fandom respectively (over the sets of users who engage with each affordance). There is little variance ($\pm$5%) in these results when using the user-centric dataset. In short, organic *Q* sessions are, in the most literal sense, the least novel whereas *A* sessions are, again in this literal sense, the most novel.

To explore the extent of this artist novelty we next apply a simple measure of redundancy, or repetition which captures a user's tendency to explore vs exploit artists within a session[9,32]. Restricting our analysis again to only consider non-fandom sessions, we compute a session's redundancy as:

$$R(S) = 1 - \frac{|\operatorname{supp} S|}{|S|} \tag{2}$$

When a user decides to exploit previously-consumed artists rather than explore new ones within a session their *R* score will be higher. To account for the fact that the upper bound of *R* is constrained by the length of a session (i.e. |*S*|) we adjust *R* to be relative to other sessions with comparable lengths. This is simply performed by computing the percentile rank of a session's *R* score relative to all other sessions in the same log length bin. After applying this adjustment we then compute per affordance average redundancy scores ($R_F$).

Considering first the affordance-centric perspective, as depicted in Fig. 2b, just as we found previously that both in terms of spatial disparity and fandom based novelty *A* sessions tend to be one of the most diverse, likewise do they tend to foster the highest artist exploration dynamics ($\langle R_A \rangle = 0.189$). At the intra-session scale, this provides strong evidence on all grounds against the claim that algorithms foster more *bubbly* artist dynamics than their organic counterparts. Similarly, even after removing sessions of the fandom kind, *Q* sessions, previously found to be the least diverse in terms of spatial disparity and fandom based novelty, likewise, exhibit the strongest exploitation dynamics ($\langle R_Q \rangle = 0.475$). If any affordance were to be deemed bubbly at the intra-session scale it would be the organic *Q* affordance.

Regarding other affordances we find that users' personal playlist (*P*) sessions are surprisingly redundant ($\langle R_P \rangle = 0.335$) despite having relatively low fandom rates and, moreover, being the most spatially diverse – although these session types typically foster at least some level of artist novelty they also encourage strong artist exploitation dynamics. We find that the inverse dynamic holds for *E* sessions, as users tend to use them in a more exploitative manner ($\langle R_E \rangle = 0.286$). They can nevertheless, on occasion, be highly artist-focused, reflected in their high fandom levels: we may simply hypothesise that this may be due to editorial playlists dedicated to a specific artist. Similarly, from the multi-affordance user-centric perspective we observe comparable differences between affordances: $\langle R_Q \rangle = 0.439$, $\langle R_P \rangle = 0.292$, $\langle R_A \rangle = 0.202$ and $\langle R_E \rangle = 0.288$.

In summary these findings suggest that, at the intra-session scale, there is some discrepancy between diversity measures which rely on discrete vs. dense item representations, particularly for the *P* affordance. Nevertheless, when considered in parallel, these perspectives offer a nuanced means of characterising and differentiating user behaviour under various levels of algorithmic, human and self-driven organic assistance. We furthermore observe a weak linear correlation between users' spatial disparity and novelty ($\langle r \rangle = 0.261$ over all affordances), confirming that the relationship users have with diversity and exploration is highly varied[17] even when users' consumptions are disaggregated by affordances.

## Inter-session diversity

Having shown that algorithmic devices may foster artist diversity rather than concentration at the intra-session scale, we now move on to explore how users' achieve inter-session diversity that is, how their sessions vary from one session to the next and how this differs under various levels of algorithmic / organic assistance. To this end, we firstly measure diversity from a spatial perspective, applying once again the *GS* score and then subsequently, from a discrete set-theoretic perspective by constructing an entropy-based monopartite projection of users' session-artist bipartite networks. At this stage of our analysis, fandom sessions are now included.

### Spatial disparity analysis

We first consider affordance-centric subsets and compute, for each affordance *F* , weighted session centroids $\mu_S$ for all $S \in S_F$ over our 1-month observation window. We subsequently compute *GS* scores over all these centroids that we denote as $GS(S_F)$ since it is computed over the collection of sessions, $S_F$. We finally consider averages over users of each affordance-centric dataset. As depicted in Table 1, this scale adds further nuance to the previous intra-session analysis. Despite *A* sessions previously being found to be the most diverse intra-session, it appears that over time, these sessions tend to become concentrated, just like the *P* and *E* kind but to a lesser degree. On the other hand, at this scale organic *Q* sessions are clear outliers with respect to how users spread their attention around different regions of a platform's informational landscape: these sessions are the most spatially diverse over time despite the fact that they lack diversity at the intra-session scale, as we previously demonstrated. Likewise we observe remarkably similar dynamics when conducting a user-centric analysis (see Supplementary Information).

All in all these findings underline the importance of aggregation scale when conducting algorithmic diversity appraisals. As we have seen for the *A* affordance, this can have a drastic effect upon the extent to which confinement dynamics are found.

### Novelty analysis

We next focus on capturing the overlap of users' sessions within each platform affordance. To this end we start by modelling a user's session-artist interactions as an undirected bipartite network whose edges reflect featurings of artists within sessions. These networks enclose co-listening patterns exhibited across sessions and are constructed per affordance: for each user in each of the 4 affordance-centric subsets, one bipartite network will be constructed for each user. In the user-centric phase of our analysis, 4 bipartite networks will be constructed per user since they engage with all 4 platform affordances.

To capture item concentration we generate monopartite projections, on the session side, of these bipartite networks, whereby two sessions are connected when they are deemed to be significantly similar in terms of the artists they share. To differentiate between session similarities which arise solely from users' artist favoritism and session lengths even in the absence of any inter-dependence between sessions we apply the Bipartite Configuration Model (BiCM)[31] as a null model. This model randomly re-configures edges between artists and sessions whilst at the same time preserving degree distributions on both the session and artist side, thus preserving a user's relative artist preference and session length. This eventually allows us to connect sessions which are *significantly* similar and compute an index $\gamma$, a set-theoretic notion of inter-session similarity which trivially captures the proportion of sessions that are connected. Further technicalities are clarified in the "Methods" section.

As presented in Table 1, when conducting an affordance-centric analysis we observe that organic *P* sessions exhibit the greatest artist overlap and thus, greater tendency to exploit the same artist sets repeatedly over time. In contrast, *Q* and *E* sessions exhibit similarly moderate overlap whilst for the *A* session this effect is even more exaggerated. Likewise this effect was also observed when conducting a user-centric analysis. We hypothesise this difference may be due to the increased stability of a user's personal playlists in comparison to algorithmic devices which, by contrast, might churn out novel item lists more frequently, rather than simply appending to existing curated artist sets (e.g., adding a song to one's own playlist).

Combined with our prior dense, spatial disparity analysis this leads us to an updated filter-bubble narrative which once again, underscores the tension between diversity measures built upon discrete and dense item representations[17] – over time music streaming platforms' algorithmic devices may promote novelty even more so than what users attain organically but nonetheless, this novelty is more semantically confined.

## Affordance similarity

Having understood that conclusions surrounding bubbly confinement are highly dependent on the scale at which diversity measurements are applied and furthermore, how diversity is framed we now move on to study inter-affordance similarity. In order to draw relative comparisons between all affordance combinations we concentrate solely on conducting a user-centric analysis (i.e. users who engage with all 4 affordances) when studying diversity in terms of novelty and spatial disparity.

### Spatial disparity analysis

For each user, we compute an overarching per-affordance centroid from each of their session centroids ($\mu_S$) constructed during our inter-session analysis. At a high level this captures a vector representation of a user's average session within each of the 4 platform affordances. To capture similarity between affordances, we calculate the cosine similarity between each centroids which we then z-normalise per user to capture relative rather than absolute inter-affordance similarities.

As shown in Table 3 which reports average normed inter-affordance similarities we find that the artist regions users exploit with the *A*, *P* and *Q* affordances are on-average the closest in vector space whilst, quite strikingly, those exploited with the *E* affordance appear to consistently reach more distant regions of vector space which most likely reflect semantically distinct artists.

| *Measure type* | **Spatial Disparity** | | | **Novelty** | | |
|---|---|---|---|---|---|---|
| | *P* | *A* | *E* | *P* | *A* | *E* |
| *Q* | 0.357 | 0.389 | -0.434 | **0.421** | 0.243 | -0.012 |
| *P* | | **0.499** | -0.490 | | 0.011 | -0.280 |
| *A* | | | -0.320 | | | -0.233 |

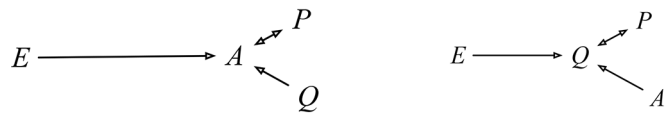


**Table 3**. Average z-normed inter-affordance similarities for each type of measures: spatial disparity (cosine similarity) vs. novelty (set-theoretic $\langle\gamma_{\cdot,\cdot}\rangle$). The subgraphs below summarize the findings by connecting each affordance to the one it is most similar to, with an arc of comparable length.

Succinctly, editorial human curated affordances appears to play a valuable role in helping users consume disparate artists styles more so than for affordances whose curations are either implicitly algorithmically personalised ($A$) or explicitly organically personalised by the user themselves ($P$, $Q$).

### Novelty analysis

Once again, we utilise the entropy-based BiCM model to construct a session network upon which we can apply the $\gamma$ measure to capture item similarity, yet not just within but now between affordances. To this end, we compute a monopartite session projection on the bipartite network that reflects all of a given user's sessions, irrespective of affordance. For each pair of affordances, we then filter to only consider nodes and edges which can be associated with the two given affordances. This yields a further session-session bipartite network whereby sessions on each side of the bipartite network all emerge from the same affordance. From this network we then compute $\gamma$ measures on each side which, in turn, reflects how item sets are mirrored between affordances. We then compute average values over both $\gamma$ computations. We eventually z-normalise all $\gamma$ scores to consider relative differences between affordances and present average z-norms per affordance pair in Table 3, where we also provide closest neighbour subgraphs which summarise affordance relative proximities.

From this perspective we now observe that there tends to be the biggest artist overlap between users' engagements within organic affordances ($P$, $Q$). Combined with the fact that the artists users consume with organic affordances tend may be relatively dissimilar from a spatial perspective –more so than say, ($P$, $A$) or ($A$,$Q$)– this suggests that these affordances may play a dual role: both reinforcing user tastes through repetition, as well as bursting users out of their own semantic bubble, presumably through organic queries ($Q$) which tend to be the most diverse at the inter-session scale. For users' $E$ sessions we observe a more consistent narrative when applying both spatial and novelty-based diversity appraisals — just as before they are consistently the most novel in comparison to other affordances. In essence, we find that at the inter-affordance scale, diversity measures built upon dense and discrete item representations, which capture spatial disparity and novelty respectively, provide a more homogeneous description of affordances' relative diversity differences, in comparison to what we observed at the previous lower aggregation scales.

## Discussion

Studying music streaming users' on-platform consumption diversities across several unique affordances which each offer variable levels of algorithmic and human assistance, the contribution of this work provides strong empirical evidence that both scale and information representations may have a marked impact upon the conclusions reached when conducting organic appraisals of algorithmic confinement. Progressing from the short-term intra-session to aggregate inter-session scale our work demonstrates that diversity differences between affordances can quickly change particularly, in the case of algorithmic affordances. Whilst music streaming platforms' algorithmic devices appear to foster diversity at the short-term, bursty session level these sessions nevertheless, over time fail to provide a substantial level of spatial disparity and thus, do not particularly appear to assist users in exploring novel regions, genres, or artist styles any more than what they attain organically.

Nothwithstanding, considering diversity measures built upon discrete (capturing novelty) and dense (capturing spatial disparity) item representations in parallel allowed us to add further nuance to this conclusion: both at the short term session scale and over time, algorithmic curation devices may promote more novelty than what users reach organically but nevertheless, this novelty is more semantically confined. Thus, we theorise that users' adoption of algorithmic devices can in some sense, break them out of their bubbles by exposing them to novelty whilst at the same time, reinforcing the boundaries of this bubble by ensuring that content is both spatially homogeneous from one session to the next and a close mirror of what they consume organically. Based on these findings we suggest that in order to reach more nuanced conclusions regarding the role of algorithmic devices in fostering bubbly confinement dynamics it may well moving forward, be fundamental to exploit several methods which leverage different aggregation scales and information representations.

Whilst we contend that this message is valuable for a broad variety of researchers studying the role of algorithmic devices on-platform it is also important to acknowledge that there are several limitations of this work. For one, we focus exclusively on the case of the Deezer music streaming platform. Whilst indeed, many of the affordances be that organic, algorithmic or editorial may be generalisable to other platform types, the inner workings of such devices and, furthermore, the interfaces they offer, may of course vary from platform to platform. Whilst so far most research performing multi-affordance organic appraisals of algorithmic confinement has tended to focus on this platform type we nevertheless suggest that a prominent direction for future work may also be found in testing the relation between different diversity measures (dense or spatial) on other platform types (e.g. social media platforms). In addition, we suggest that although outside the scope of this work future research should also focus on bridging the gap between $A$ exposure and multi-affordance literature thereby acknowledging that an algorithmic consumption is at least in part also mediated by an organic human selection process.

## Methods

### SVD based feature space

We construct a co-occurrence matrix $M$ based on intra-monadic session co-occurrences whereby rows define the artist structure to be embedded. We perform an over-sampling of affordances which are in minority classes, thus balancing the number of session per affordance. We proceed to populate $M$ with within-session co-occurrence counts. We then cast $M$ to a positive point wise mutual information matrix to account for co-consumption dynamics which may emerge from popularity effects alone. Finally, we apply Truncated Singular Value Decomposition (SVD) to the matrix to generate a low-dimensional ($k = 128$) embedding space. More

formally, SVD is a factorisation technique which seeks to approximate a matrix $M \in R^{n \times m}$ as the product of three matrices, $U \in \mathbb{R}^{n \times k}$, $\Sigma \in \text{diag}\left(\mathbb{R}_{+}^{k}\right)$ and $V \in \mathbb{R}^{m \times k}$. In our analysis we make use of the $U\Sigma$ component as our latent item embedding which from hereon out we shall denote as $L$, an $n \times k$ matrix whose rows (i.e. $l_1, l_2, ...l_n$) correspond to the latent representations of artists. For simplicity let artist ids exist in the range $[1 \ldots n]$ and let an artist with id *i* have the latent vector representation $l_i$.

## GS score

We estimate spatial disparity through an activity-weighted so-called "Generalist-Specialist", or GS, score[2,28]. Let *S* represent the list of artist ids associated with a given user's session. Let us first define the weighted centroid of a session in vector space as:

$$\mu_S = \frac{\sum_{j \in S} w_j l_j}{|S|} \tag{3}$$

where $l_j$ denotes the latent representation of artist *j* and $w_j$ denotes the number of plays associated with them within *S* (note that $\sum_j w_j = |S|$). To capture intra-session spatial disparity we compute the *GS* score of *S* as:

$$GS\left(S\right) = \frac{1}{|S|} \sum_{j \in S} w_j \frac{l_j \cdot \mu_S}{\|l_j\| \cdot \|\mu_S\|} \tag{4}$$

Similarly, we compute spatial disparity at the inter-session scale as:

$$GS\left(S_F\right) = \frac{1}{|S_F|} \sum_{S \in S_F} \frac{\mu_S \cdot \mu_F}{\|\mu_S\| \cdot \|\mu_F\|} \tag{5}$$

whereby for a given user and their monadic consumptions, $S_F$ reflects the family of sessions associated with affordance $F \in \{A, E, P, Q\}$ and $\mu_F$ denotes the average location of *F*-session centroids in vector space.

For each subset (affordance-centric or user-centric) we only report *GS* scores for sessions that contain $< 20\%$ niche artists whereby an artist is defined as niche if they have been listened to by $\leq 10$ users. In the case where a session contains more than one niche artist but still the session is composed of $< 20\%$ niche artists we simply drop the artists classified as niche from the set. This has a minimal effect on user sessions due to the long-tailed nature of popularity on-platform: on average across each affordance-centric user set $5.76\%$ of a user's average session is composed of "niche" artists. This ensures that we filter artists whose structure in *M*, and thus also in vector space, is poorly represented, thereby avoiding deriving spatial concentration conclusions which may have more to do with popularity than underlying semantics.

As detailed in the Supplementary Information, we attain equivalent relative disparity differences between affordances when measuring within-session disparity with both normalised genre entropy and the GS score therefore, suggesting that our results would likely be the same if a genre-based item representation was applied throughout.

## Bipartite configuration model (BiCM)

We use BiCM[31] as a baseline to compute item diversity in the inter-session and inter-affordance cases i.e., where the significance of overlaps between consumption sets needs to be computed. BiCM interprets links as independent random variables and, as such, defines a grandcanonical probability measure (i.e. over all possible network configurations which preserve node degrees on both sides). To illustrate on inter-session diversity, and following the methodology defined in[33], we conceptualise session similarity using so-called *V*-motifs which reflect the number of shared artists between any two sessions and make use of the BiCM model to define significance of *V*-motifs w.r.t. to the expectation under the BiCM null model ($\alpha = 0.05$). (Solver convergence rates are very high – for each of the affordance-centric sets we observe that the solvers tested were on average able to converge on 94.46% of users tested. In cases of repeated non-convergence these users are dropped from the analysis set). For further details of methodological technicalities we refer readers to the work of[31,33].

We may now measure session homogeneity, in item terms, using each of the *n* projected session networks $\mathcal{S}_u$ of a user *u* engaging in *n* affordances. We constuct an adjusted overlap measure $\gamma$ which trivially captures the proportion of sessions $S_i \in \mathcal{S}_u$ that are connected, formally defined as:

$$\gamma = |\{S_i \in \mathcal{S}_u \text{ s.t. } \deg(S_i) > 0\}| / |\mathcal{S}_u| \tag{6}$$

An illustration of this pipeline is shown in Fig. 3. We proceed similarly to compute inter-affordance similarity, except that we configure affordance-based pseudo-sessions gathering all listening events associated with a given affordance. We then compute inter-pseudo-session similarity in the same way as before on pairs of affordance-based pseudo-sessions, which actually reflects a measure on the similarity between each pair of considered affordances.

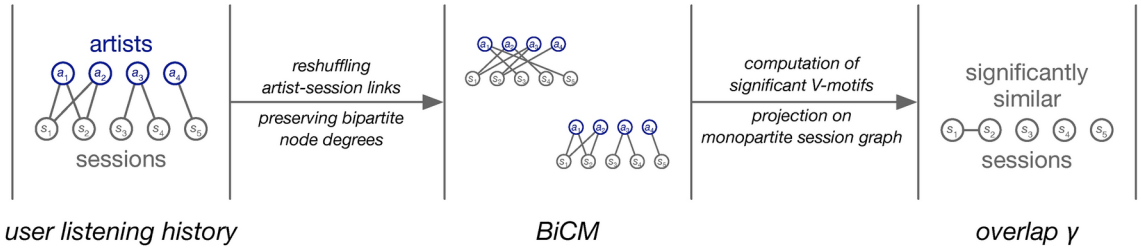


**Fig. 3**. Schematic illustration of the BiCM session overlap computation pipeline, from an empirical bipartite session-artist network to the computation of significant V-motifs or monopartite projection links i.e., significant connections between sessions in terms of their artist overlap.

| Affordance | Sub-affordance |
|---|---|
| P | playlist_page |
| Q | album_page, playlist_radio, shuffled_collection, artist_top, profile_top, suggest_track, search_page_track, |
| | recently_played_playlist, profile_playlists, suggest_playlist, dynamic_page_album, dynamic_page_playlist, search_page, |
| | history_page, profile_artists, shuffled_offline, profile_albums, recently_played_user_radio, recently_played_album, |
| | cached_content, suggest_album, siri_album, artist_randomdiscography, siri_track, search_playlist, siri_artist_top, |
| | widget_album, artist_discography_subpage, siri_playlist, recently_played_user, recently_played_radio, widget_artist, |
| | suggest_radio, artist_discography,siri_radio, siri_artist_radio, player_default_playlist, search_album, |
| | artist_highlight_album, siri_shortcuts_shuffled_collection, album_discography, dynamic_page_track, search_show_episode, |
| | artist_similar_playlist, suggest_artist, menu_panel_notification_album, personalsong_page, search_artist, profile_radios, |
| | suggest_livestream, suggest_show_episode, artist_highlight_playlist, artist_related_playlists_subpage, recently_played_artist |
| A | feed_user_radio, dynamic_page_user_radio, profile_user_radio, widget_flow, dynamic_page_smarttracklist, |
| | dynamic_page_artist_radio, flow_tab_smarttracklist, track_mix_contextual, siri_flow, dynamic_page_radio, |
| | artist_smartradio, widget_smarttracklist, smarttracklist_page, siri_shortcuts_user_radio, flow_tab_flow, radio_page, |
| | feed_playlist, dynamic_page_livestream, feed_track, track_mix_30s, smartradio_page |
| E | dynamic_page_playlist, playlist_page, playlist_radio |

**Table 4**. Classification scheme used to derive affordance categories on Deezer.

## Affordance nomenclature

For full transparency the classification scheme used to classify affordances is depicted in Table 4. Listening events whose affordance is tagged as "external" (i.e. access was achieved off-platform for instance, via an API) are filtered since we are interested in studying on-platform rather than off-platform confinement.

## Filtering strategies

To account for the variation of monadic session types (in terms of affordances) across users we introduce two user filtering strategies filtering strategy which provide an affordance- and user-centric perspective respectively. The former constructs one set of users per affordance $F \in (A, E, P, Q)$, first filtering sessions which do not emerge from the affordance associated with the set and then next, filtering users that have $< 3$ sessions within this set. This results in 4 sets of users who actively engage with each of the affordances in question. The sets have the following respective user populations $n_A = 12,411$, $n_E = 10,923$, $n_P = 28,896$ and $n_Q = 36,004$. The latter filtering strategy which provides a user-centric perspective imposes the more strict constraint that users must have $\geq 3$ session in each affordance and thus, constructs a set of so-called multi-affordance users. It is meant to construct a reference user set to sense-check the conclusions derived at the session level thus, reducing the risk of ecological fallacy. Applying this filering yields a user set of size $n = 1672$.

## Data availability

To elicit reproducibility we publicly release, alongside this work, the dataset used to derive all user-centric diversity conclusions. More precisely, this dataset is restricted firstly to streams which were enacted within monadic sessions (i.e. those that only use one affordance) and subsequently to users who engage with $\geq 3$ sessions in each of the Deezer platform's 4 affordances thus, focusing on "multi-affordance" users. We hope that the release of such a dataset will allow more researchers to perform organic appraisals of algorithmic confinement using multi-affordance usage logs which, to date, remain scarce due to these type of datasets generally being exclusively reserved for those working in industry partnerships. We also release the code needed to reproduce all experiments with the user-centric dataset. Access to the code and dataset can be found at: https://github.com/deshiiii/SciReports-FilterBubble/tree/main .

Received: 29 April 2024; Accepted: 9 October 2024
Published online: 03 February 2025

## References

1. Pariser, E. *The Filter Bubble: What the Internet is Hiding From You* (Penguin Press, 2011).
2. Anderson, A., Maystre, L., Anderson, I., Mehrotra, R. & Lalmas, M. Algorithmic effects on the diversity of consumption on spotify. In *Proceedings of The Web Conference 2020*, WWW '20, 2155-2165, https://doi.org/10.1145/3366423.3380281 (Association for Computing Machinery, 2020).
3. Bandy, J. & Diakopoulos, N. Auditing news curation systems: A case study examining algorithmic and editorial logic in apple news. In *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 14, 36–47 (2020).
4. Robertson, R. E. *et al.* Users choose to engage with more partisan news than they are exposed to on google search. *Nature* 1–7 (2023).
5. Bakshy, E., Messing, S. & Adamic, L. A. Exposure to ideologically diverse news and opinion on facebook. *Science* **348**, 1130–1132. https://doi.org/10.1126/science.aaa1160 (2015).
6. Cardenal, A. S., Aguilar-Paredes, C., Galais, C. & Pérez-Montoro, M. Digital technologies and selective exposure: How choice and filter bubbles shape news media exposure. *Int. J. Press/Polit.* **24**, 465–486 (2019).
7. Möller, J., Trilling, D., Helberger, N. & van Es, B. Do not blame it on the algorithm: an empirical assessment of multiple recommender systems and their impact on content diversity. *Inf. Commun. Soc.* **21**, 959–977. https://doi.org/10.1080/1369118X.2018.1444076 (2018).
8. Epps-Darling, A. C., Cramer, H. & Bouyer, R. T. Artist gender representation in music streaming. In *Proceedings of the 21st International Society for Music Information Retrieval Conference*, 248–254, https://doi.org/10.5281/zenodo.4245416 (ISMIR, 2020).
9. Villermet, Q., Poiroux, J., Moussallam, M., Louail, T. & Roth, C. *Follow the Guides: Disentangling Human and Algorithmic Curation in Online Music Consumption, 380–389* (Association for Computing Machinery, 2021).
10. Zhou, R., Khemmarat, S. & Gao, L. The impact of youtube recommendation system on video views. In *Proceedings of the 10th ACM SIGCOMM Conference on Internet Measurement*, IMC '10, 404-410, https://doi.org/10.1145/1879141.1879193 (Association for Computing Machinery, 2010).
11. Fortunato, S., Flammini, A., Menczer, F. & Vespignani, A. Topical interests and the mitigation of search engine bias. *Proc. Natl. Acad. Sci.* **103**, 12684–12689. https://doi.org/10.1073/pnas.0605525103 (2006).
12. Aiello, L. M. & Barbieri, N. Evolution of ego-networks in social media with link recommendations. *CoRR***abs/1702.01398** (2017). arxiv:1702.01398.
13. Nguyen, T. T., Hui, P.-M., Harper, F. M., Terveen, L. & Konstan, J. A. Exploring the filter bubble: the effect of using recommender systems on content diversity. In *Proceedings of the 23rd International Conference on World Wide Web*, WWW '14, 677-686, https://doi.org/10.1145/2566486.2568012 (Association for Computing Machinery, 2014).
14. Datta, H., Knox, G. & Bronnenberg, B. J. Changing their tune: How consumers' adoption of online streaming affects music consumption and discovery. *Mark. Sci.* **37**, 5–21 (2018).
15. Haim, M., Graefe, A. & Brosius, H.-B. Burst of the filter bubble? effects of personalization on the diversity of google news. *Digit. J.* **6**, 330–343 (2018).
16. Beuscart, J.-S., Coavoux, S., Maillard, S. & Carey-Libbrecht, L. Music recommendation algorithms and listener autonomy. *Reseaux* **213**, 17–47 (2019).
17. Mok, L., Way, S. F., Maystre, L. & Anderson, A. The dynamics of exploration on spotify. In *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 16, 663–674. https://doi.org/10.1609/icwsm.v16i1.19324 (2022).
18. Jürgens, P. & Stark, B. Mapping exposure diversity: The divergent effects of algorithmic curation on news consumption. *J. Commun.* **72**, 322–344 (2022).
19. Katz, E., Lazarsfeld, P. F. & Roper, E. *Personal Influence: The Part Played by People in the Flow of Mass Communications* (Routledge, 2017).
20. Davis, J. L. *How Artifacts Afford: The Power and Politics of Everyday Things* (MIT Press, 2020).
21. Roth, C., Mazières, A. & Menezes, T. Tubes and bubbles—topological confinement of youtube recommendations. *PLoS ONE* **15**, e0231703 (2020).
22. Huszár, F. et al. Algorithmic amplification of politics on twitter. *Proc. Natl. Acad. Sci.* **119**, e2025334119 (2022).
23. Wang, S. et al. A survey on session-based recommender systems. *ACM Comput. Surv. (CSUR)* **54**, 1–38 (2021).
24. Hosey, C., Vujović, L., St. Thomas, B., Garcia-Gathright, J. & Thom, J. Just give me what i want: How people use and evaluate music search. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12 (2019).
25. Li, A. *et al.* Search mindsets: Understanding focused and non-focused information seeking in music search. In *The World Wide Web Conference*, WWW '19, 2971-2977, https://doi.org/10.1145/3308558.3313627 (Association for Computing Machinery, 2019).
26. Afchar, D., Hennequin, R. & Guigue, V. Of spiky SVDs and music recommendation (2023). arxiv:2307.01212.
27. Levy, O. & Goldberg, Y. Neural word embedding as implicit matrix factorization. In *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2*, NIPS'14, 2177-2185 (MIT Press, 2014).
28. Waller, I. & Anderson, A. Generalists and specialists: Using community embeddings to quantify activity diversity in online platforms. In *The World Wide Web Conference*, WWW '19, 1954-1964, https://doi.org/10.1145/3308558.3313729 (Association for Computing Machinery, 2019).
29. Wang, Y., Huang, H., Rudin, C. & Shaposhnik, Y. Understanding how dimension reduction tools work: an empirical approach to deciphering t-sne, umap, trimap, and pacmap for data visualization. *J. Mach. Learn. Res.* **22**, 9129–9201 (2021).
30. Lapesa, G., Evert, S. & Schulte im Walde, S. Contrasting syntagmatic and paradigmatic relations: Insights from distributional semantic models. In *Proc. Third Joint Conf. on Lexical and Computational Semantics (*SEM 2014)*, 160–170 (Association for Computational Linguistics and Dublin City University, 2014).
31. Saracco, F., Clemente, R. D., Gabrielli, A. & Squartini, T. Randomizing bipartite networks: the case of the world trade web. *Sci. Rep.*[SPACE]https://doi.org/10.1038/srep10595 (2015).
32. Louail, T. & Barthelemy, M. Headphones on the wire. *arXiv preprint*[SPACE]arXiv:1704.05815 (2017).
33. Saracco, F. et al. Inferring monopartite projections of bipartite networks: an entropy-based approach. *New J. Phys.* **19**, 053022. https://doi.org/10.1088/1367-2630/aa6b38 (2017).

## Acknowledgements

This paper has been partially realized in the framework of the "RECORDS" grant (ANR-2019-CE38-0013) funded by the ANR (French National Agency of Research)

## Author contributions

D.S, V.C. and C.R. designed the study. D.S performed the calculations. D.S. designed the computational framework and analysed the data. D.S. took the lead in writing the manuscript. D.S. and C.R. wrote the paper. C.R. supervised the project.

## Declarations

### Competing Interests

The authors declare no competing interests.

### Additional information

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1038/s41598-024-75967-0.

**Correspondence** and requests for materials should be addressed to D.S.

**Reprints and permissions information** is available at www.nature.com/reprints.

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025